Predicting self-reported heart disease in the 2015 BRFSS

CS-433 Project 1

Abstract—We predict whether an adult in the 2015 Behavioral Risk Factor Surveillance System has been told they had a heart attack or coronary heart disease. Of 328,135 labelled rows, 8.83% are positive, so accuracy is a weak target: an all-negative rule already scores 0.912. Survey codes were treated as categories, items that leak the label were removed, and a weighted ℓ_2 logistic regression was fit on the 300 one-hot columns with the largest mutual information. Five stratified folds give class-1 F1 0.427 ± 0.002. Reading the same codes as numbers, or keeping only missingness flags, lowers F1. The file submissions/onehot_mi_top.csv is this model. Its public test score is not yet known.

I. Problem and data

The label is the BRFSS derived item for myocardial infarction, angina, or coronary heart disease, stored as $\{-1, +1\}$. The hidden test file has 109,379 rows and the same 321 columns. The competition scores class-1 F1, so that is the selection metric. Precision-recall AUC is reported beside it because it does not depend on the cutoff.

The positive rate is not uniform. It is about 1% under age 45 and about 28% for men aged 75 to 80. One score and one cutoff have to serve both groups. Train and validation F1 on the linear models differed by about 0.01, so the limit is overlap of the classes, not overfitting of a particular solver.

II. Feature decisions

Columns that record the survey rather than the person were dropped before any importance score could put them back. These include state, interview date, sequence number, and phone-sample checks. Do-not-know and refused codes were set to missing: 7 and 9 on yes/no items, and 77 and 99 on day counts. On day counts, 88 means zero days and was recoded to 0. Body-mass index above 60 was capped. Four items answered only after the event, HAREHAB1, STREHAB1, CVDASPRN, and ASPUNSAF, were removed so the model cannot see the outcome.

Missing entries are informative. For several clinical items the positive rate among blank rows differs from the rate among answered rows, and the missingness indicator itself ranks above chance. A missing flag is therefore kept when a recoded column is blank on at least 5% of fit rows, and always for income, cholesterol, diet, seat belt, strength training, and the HIV test. Flags are not enough on their own.

Two representations were compared with the model held fixed. The codebook path recodes each item to a clinical yes/no or an ordered code, fills the mode, standardizes, and adds the squares of age and body-mass index plus products of age with sex, smoking, diabetes, general health, and cholesterol. Mode fill beat median and mean on an earlier single split (validation F1 0.418, 0.417, and 0.416). The one-hot path does not trust the numeric codes. A value held by at least 1% of the fit rows becomes its own column. Rarer values are pooled or cut into at most five quantile bins, and a common missing pattern is its own level. Constant and duplicate columns are dropped. Mutual information with the label then keeps 300 columns, which are standardized. Levels, scales, the duplicate check, and the ranking use the fit rows only.

III. Model and protocol

The classifier is logistic regression. Let $y_i \in \{0, 1\}$ and let row i have weight w_i, with the positive rows together weighing twice the negative rows. The bias is not penalized. With penalty $\lambda = 10^{-5}$,

$$\min_{\theta} \ \frac{1}{n} \sum_{i=1}^{n} w_i \, \ell(y_i, \theta_0 + x_i^{\top} \theta_{1:}) + \lambda \|\theta_{1:}\|_2^2. \qquad (1)$$

The loss ℓ is the logistic negative log-likelihood. Weights are fit by 300 steps of FISTA. A row is predicted $+1$ when its score is at least a cutoff t. Inside each outer training fold, 15% of the rows, stratified, are held out and t is the value that maximizes class-1 F1 there. The outer fold is used once, for the reported metrics. There are five folds, seed 42. For the submission, the same 85/15 split is applied to every labelled row. The test matrix is used only at prediction time. On that fit, $t = 1.75$ and 13.16% of test rows are called positive.

The six course solvers were compared earlier on the codebook design. Least squares, ridge, gradient descent, stochastic gradient descent, and both logistic regressions landed between validation F1 0.421 and 0.430 on an age split. The solver was not the gap.

IV. Results

Table I changes only the preprocessing. The submitted row is onehot_mi_top. It wins class-1 F1 on every fold

Table I
Five-fold means. The standard deviation of F1 is 0.001 to 0.003.

Pipeline	F1	PR-AUC	ROC	cols
one-hot, top 300	0.427	0.388	0.861	300
one-hot, ℓ_1	0.425	0.385	0.859	1753
one-hot, all levels	0.422	0.382	0.857	1753
codebook	0.422	0.386	0.859	171
codebook plus one-hot	0.422	0.382	0.857	1924
one-hot $\times$ age ≥ 65	0.422	0.381	0.857	1864
median and all flags	0.410	0.359	0.851	464
median, low missing	0.407	0.355	0.850	215
flags, top 40 sources	0.395	0.336	0.841	227
missing flags only	0.343	0.270	0.807	187

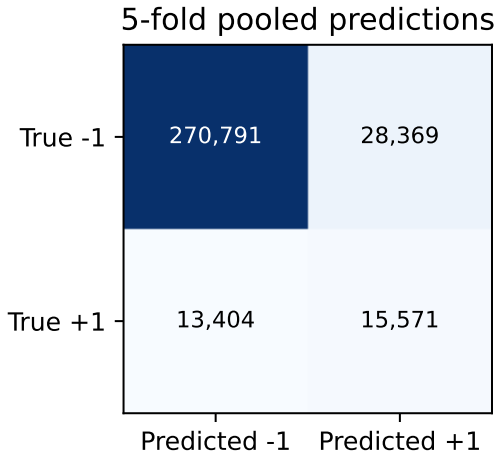


Figure 1. Pooled predictions of the submitted model. Each cell is a count over the five test folds.

against the codebook, by about 0.005, which is larger than the fold standard deviation of 0.002. The precision-recall gap against the codebook is smaller than the fold noise, so the two representations are close as rankers and the gain is the cutoff.

Figure 1 pools the five test folds, so each labelled row is scored once. There are 15,571 true positives, 28,369 false positives, 13,404 false negatives, and 270,791 true negatives. Pooled precision is 0.354, recall is 0.537, F1 is 0.427, and accuracy is 0.873. Most errors are false alarms. Of 28,975 positive rows, 13,404 are missed.

On a separate 80/20 split of the codebook features, two logistic models split at age 65 scored validation F1 0.430, against 0.425 for one model. A sex split scored 0.427 combined, and four sex-by-age models scored 0.429. Those splits are not in Table I and are not subtracted from 0.427. Changing the positive-class cost from 0.5 to 6 moved the age-split F1 by 0.001: the cutoff slides back and cancels the weight. An F2 cutoff, with recall near 0.80 and precision near 0.25, dropped F1 to about 0.38 and was rejected because the leaderboard uses F1. An earlier public submission, from a weaker preprocessing, scored 0.391. The 0.435 F1 on the 15% slice that chose t for the new file is optimistic and is not the estimate.

V. Why the other attempts lost

Reading codes as numbers treats 7, 9, and 88 as magnitudes. Median imputation plus missing flags therefore drops F1 by 0.017 and precision-recall AUC by 0.029 relative to the selected one-hot model. Restricting that path to columns with under 50% missing does not repair it. Flags on the 40 columns with the highest mutual information still leave the answered values as numbers, and F1 falls to 0.395. Flags alone keep the missingness signal and discard the answers, which costs 0.084 F1. The signal in a blank is real, and it is smaller than the signal in the answer.

Using every one-hot level, about 1,750 columns, scores 0.422. Rare levels add columns that the 300-column ranking had already discarded. An ℓ_1 penalty of $2 \cdot 10^{-4}$ on that full matrix leaves about 1,200 nonzero weights and F1 0.425, still below the mutual-information selection, and the fit is about three times slower. Products of the one-hot columns with an age-at-least-65 flag, and the union of the codebook matrix with the full one-hot matrix, both sit at 0.422. The added columns repeat information the selected levels already carry. Permutation importance on the codebook fold points to age, general health, prior stroke, sex, and cholesterol. On the full one-hot fold the largest drops are sex, stroke, high cholesterol, and general health. Age is split across many level columns there, so one raw age column looks less important even though the levels remain in the model.

A single linear score can change one feature's slope only where a product column exists. A gradient-boosted tree or a neural net would add those changes without writing each product by hand. No finished cross-validation of either model is in this working copy, so no F1 is reported for them. The linear comparisons above are the ones that can be reproduced from this code.

VI. Conclusion

The submitted model is weighted ℓ_2 logistic regression on 300 one-hot levels chosen by mutual information, with the class-1 F1 cutoff chosen on held-out rows. The local estimate is F1 0.427, precision 0.354, and recall 0.537. Preprocessing moved the score more than the choice of linear solver, the class weight, or an age or sex split. The public score of onehot_mi_top.csv is still unknown and should be compared with this estimate rather than with the cutoff-selection slice.